

ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination questions
January 2019

Instructions: Closed book. NO CALCULATORS are allowed. This examination contains **seven** problems, each worth 20 points. Do any **five** of the seven problems. Show your work. Clearly indicate which **five** are to be graded.

PLEASE GRADE PROBLEMS: 1 2 3 4 5 6 7

1. Assume that $fl(x)$ is a floating point representation of a real number x with relative error bounded above by $\delta > 0$.

(a) Show that $\left| \frac{\log(1-\delta)}{\log x} \right|$ is an upper bound on the relative error of $\log(fl(x))$.

(b) Find the condition number of $\log x$ and explain the connection to part (a).

2. Show that for $n \geq 0$, the sequence

$$x_{n+1} = \frac{1}{2}x_n \left(1 + \frac{a}{x_n^2} \right)$$

has convergence of second order to the root $\sqrt{a}$.

3. Solve the following system using Cholesky factorization:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 29 & 41 \\ 3 & 41 & 179 \end{pmatrix} \vec{x} = \begin{pmatrix} 40 \\ 510 \\ 1690 \end{pmatrix}$$

4. Gauss Quadrature

(a) Derive the two point Gaussian integration formula for

$$I(f) = \int_0^1 f(x) dx.$$

(b) Evaluate the integral using the two-point Gauss Quadrature formula derived:

$$\int_0^1 \frac{dx}{5x+1}.$$

SOLUCIÓN — PROBLEMA 1

Problema 1 — Error relativo de $\log(fl(x))$ y condición de $\log$

(a) Por definición, $fl(x) = x(1 + \epsilon)$ con $|\epsilon| \leq \delta$. Entonces

$$\log(fl(x)) = \log x + \log(1 + \epsilon).$$

El error relativo de $\log(fl(x))$ respecto a $\log x$ es

$$\text{RE} = \frac{|\log(fl(x)) - \log x|}{|\log x|} = \frac{|\log(1 + \epsilon)|}{|\log x|}.$$

Como $\log$ es estrictamente creciente, para $\epsilon \in [-\delta, \delta]$,

$$\log(1 - \delta) \leq \log(1 + \epsilon) \leq \log(1 + \delta).$$

Necesitamos acotar $|\log(1 + \epsilon)|$. Para $\epsilon \geq 0$: $|\log(1 + \epsilon)| = \log(1 + \epsilon) \leq \log(1 + \delta)$.
Para $\epsilon < 0$: $|\log(1 + \epsilon)| = -\log(1 + \epsilon) \leq -\log(1 - \delta) = |\log(1 - \delta)|$.

Comparamos las dos cotas $\log(1 + \delta)$ y $|\log(1 - \delta)| = \log \frac{1}{1 - \delta}$:

$$(1 + \delta)(1 - \delta) = 1 - \delta^2 \leq 1 \implies 1 + \delta \leq \frac{1}{1 - \delta} \implies \log(1 + \delta) \leq \log \frac{1}{1 - \delta} = |\log(1 - \delta)|.$$

Así, la cota más grande (el peor caso) es $|\log(1 - \delta)|$, alcanzada en $\epsilon = -\delta$. Por lo tanto, para todo $\epsilon \in [-\delta, \delta]$,

$$|\log(1 + \epsilon)| \leq |\log(1 - \delta)| \implies \text{RE} = \frac{|\log(1 + \epsilon)|}{|\log x|} \leq \left| \frac{\log(1 - \delta)}{\log x} \right|. \blacksquare$$

(b) El número de condición (relativo) de $f(x) = \log x$ es

$$\kappa(\log, x) = \left| \frac{x f'(x)}{f(x)} \right| = \left| \frac{x \cdot \frac{1}{x}}{\log x} \right| = \frac{1}{|\log x|}.$$

Conexión con (a): para δ pequeño, $\log(1 - \delta) = -\delta + O(\delta^2)$ (serie de Taylor de $\log(1 - \delta)$ en $\delta = 0$), así que la cota de (a) es

$$\left| \frac{\log(1 - \delta)}{\log x} \right| = \frac{\delta + O(\delta^2)}{|\log x|} = \kappa(\log, x) \cdot \delta + O(\delta^2).$$

Esto es exactamente la fórmula de propagación de primer orden del error relativo: $\text{error relativo de } f(x) \approx \kappa(f, x) \cdot \text{error relativo de } x$, confirmando que la cota exacta de (a) recupera, en el límite $\delta \rightarrow 0$, el número de condición estándar $\kappa(\log, x) = 1/|\log x|$.

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Problema 2 — Orden de convergencia de la iteración para $\sqrt{a}$

La iteración dada, $x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$, es precisamente el método de Newton aplicado a $f(x) = x^2 - a$ (comprobación: $x - \frac{x^2 - a}{2x} = \frac{x^2 - x^2 + a}{2x} = \frac{a}{2x}$).

Sea $e_n = x_n - \sqrt{a}$. Entonces

$$x_{n+1} - \sqrt{a} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) - \sqrt{a} = \frac{x_n^2 - 2\sqrt{a}x_n + a}{2x_n} = \frac{(x_n - \sqrt{a})^2}{2x_n} = \frac{e_n^2}{2x_n}.$$

Si $x_n \rightarrow \sqrt{a} > 0$ (con $a > 0$, $x_n \neq 0$ en el proceso), entonces

$$\frac{e_{n+1}}{e_n^2} = \frac{1}{2x_n} \xrightarrow{n \rightarrow \infty} \frac{1}{2\sqrt{a}},$$

límite finito y no nulo. Por definición, esto es **convergencia de segundo orden** (cuadrática) hacia $\sqrt{a}$, con constante asintótica $C = \frac{1}{2\sqrt{a}}$. ■

Verificación: con $a = 4$, $x_0 = 1$: $x_1 = \frac{1}{2}(1 + 4) = 2.5$, $e_1 = 0.5$; $x_2 = \frac{1}{2}(2.5 + 1.6) = 2.05$, $e_2 = 0.05$; $e_2/e_1^2 = 0.05/0.25 = 0.2 = 1/(2 \cdot 2)$ ✓ (coincide con $1/(2\sqrt{4}) = 1/4$... en el límite; aquí $x_1 = 2.5 \neq 2$ todavía, por eso el cociente exacto es $1/(2x_1) = 0.2$, consistente con la fórmula exacta $e_{n+1}/e_n^2 = 1/(2x_n)$ para cada n , no solo en el límite).

ENUNCIADO — PROBLEMA 3

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Problema 3 — Sistema lineal vía factorización de Cholesky

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 29 & 41 \\ 3 & 41 & 179 \end{pmatrix}, \quad b = \begin{pmatrix} 40 \\ 510 \\ 1690 \end{pmatrix}.$$

Factorización $A = LL^T$. Con $L = (l_{ij})$ triangular inferior:

$$\begin{aligned} l_{11} &= \sqrt{a_{11}} = \sqrt{1} = 1; & l_{21} &= \frac{a_{21}}{l_{11}} = \frac{2}{1} = 2; & l_{31} &= \frac{a_{31}}{l_{11}} = \frac{3}{1} = 3; \\ l_{22} &= \sqrt{a_{22} - l_{21}^2} = \sqrt{29 - 4} = \sqrt{25} = 5; & l_{32} &= \frac{a_{32} - l_{31}l_{21}}{l_{22}} = \frac{41 - 3 \cdot 2}{5} = \frac{35}{5} = 7; \\ l_{33} &= \sqrt{a_{33} - l_{31}^2 - l_{32}^2} = \sqrt{179 - 9 - 49} = \sqrt{121} = 11. \end{aligned}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 5 & 0 \\ 3 & 7 & 11 \end{pmatrix}.$$

Verificación de $LL^T = A$: fila 3 de LL^T : $(3 \cdot 1, \quad 3 \cdot 2 + 7 \cdot 5, \quad 3 \cdot 3 + 7 \cdot 7 + 11 \cdot 11) = (3, \quad 6 + 35, \quad 9 + 49 + 121) = (3, 41, 179)$ ✓ (y análogamente las filas 1,2 reproducen A).

Paso 1: resolver $Ly = b$ (sustitución hacia adelante).

$$y_1 = \frac{40}{1} = 40; \quad 2(40) + 5y_2 = 510 \Rightarrow 5y_2 = 430 \Rightarrow y_2 = 86;$$

$$3(40) + 7(86) + 11y_3 = 1690 \Rightarrow 120 + 602 + 11y_3 = 1690 \Rightarrow 11y_3 = 968 \Rightarrow y_3 = 88.$$

****Paso 2: resolver $L^T x = y$ (sustitución hacia atrás), con $L^T = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & 11 \end{pmatrix}$.****

$$x_3 = \frac{88}{11} = 8; \quad 5x_2 + 7(8) = 86 \Rightarrow 5x_2 = 30 \Rightarrow x_2 = 6;$$

$$x_1 + 2(6) + 3(8) = 40 \Rightarrow x_1 + 12 + 24 = 40 \Rightarrow x_1 = 4.$$

$$\boxed{\vec{x} = (4, 6, 8)^T.}$$

Verificación directa en $A\vec{x} = b$:

- Fila 1: $1(4) + 2(6) + 3(8) = 4 + 12 + 24 = 40$ ✓
- Fila 2: $2(4) + 29(6) + 41(8) = 8 + 174 + 328 = 510$ ✓
- Fila 3: $3(4) + 41(6) + 179(8) = 12 + 246 + 1432 = 1690$ ✓

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5. Consider the two-point boundary value problem:

$$-u'' + \exp(x)u = \tan(x) \quad \text{in } (0, 1), \quad \text{with } u(0) = u(1) = 0. \quad (1)$$

- (a) Write the finite difference approximation of (1) using *centered* differences on a uniform mesh with meshsize h . Write the matrix of the system and examine how the equations would change if $u(1) = \alpha \neq 0$.
- (b) Set up the finite element approximation for this problem, based on piecewise linear elements in equidistant points. State the convergence rate in an appropriate norm.
6. Consider the matrix

$$\begin{pmatrix} \sqrt{3} & 0 & -\sqrt{3} \\ 0 & a & 0 \\ \sqrt{3} & 0 & \sqrt{3} \end{pmatrix}$$

Explain why normalized power iteration fails to converge for almost every initial vector in $\mathbb{R}^3$ when $a = 1$. For what values of a will normalized power iteration converge for almost every vector in $\mathbb{R}^3$? What will it converge to?

7. Let h be the step size used for interpolating the function $f(x) = \sin(x)$ in $\left[0, \frac{\pi}{4}\right]$ and let $p(x)$ be the quadratic (three nodes) Lagrange interpolating polynomial using equally spaced nodes. Show that $|f(x) - p(x)| \leq \frac{h^3}{9\sqrt{3}}$.

SOLUCIÓN — PROBLEMA 4

Problema 4 — Cuadratura gaussiana de 2 puntos en $[0, 1]$

(a) Mapeamos $[-1, 1] \rightarrow [0, 1]$ mediante $x = \frac{t+1}{2}$, $dx = \frac{dt}{2}$. La regla estándar de 2 puntos en $[-1, 1]$ usa nodos $t = \mp 1/\sqrt{3}$ con pesos 1 cada uno (Problema 2 del examen de agosto de 2018). Entonces

$$\int_0^1 f(x) dx = \frac{1}{2} \int_{-1}^1 f\left(\frac{t+1}{2}\right) dt \approx \frac{1}{2} \left[f\left(\frac{1-1/\sqrt{3}}{2}\right) + f\left(\frac{1+1/\sqrt{3}}{2}\right) \right].$$

Simplificando los nodos:

$$x_1 = \frac{1}{2} - \frac{1}{2\sqrt{3}} = \frac{3-\sqrt{3}}{6}, \quad x_2 = \frac{1}{2} + \frac{1}{2\sqrt{3}} = \frac{3+\sqrt{3}}{6},$$

$$\boxed{\int_0^1 f(x) dx \approx \frac{1}{2} \left[f\left(\frac{3-\sqrt{3}}{6}\right) + f\left(\frac{3+\sqrt{3}}{6}\right) \right]}.$$

Esta regla es exacta para polinomios de grado ≤ 3 .

(b) Evaluamos $I = \int_0^1 \frac{dx}{5x+1}$ con $f(x) = 1/(5x+1)$.

$$5x_1 + 1 = \frac{5(3-\sqrt{3})}{6} + 1 = \frac{15-5\sqrt{3}+6}{6} = \frac{21-5\sqrt{3}}{6} \Rightarrow f(x_1) = \frac{6}{21-5\sqrt{3}},$$

$$5x_2 + 1 = \frac{21+5\sqrt{3}}{6} \Rightarrow f(x_2) = \frac{6}{21+5\sqrt{3}}.$$

Sumando (racionalizando con diferencia de cuadrados, $21^2 - (5\sqrt{3})^2 = 441 - 75 = 366$):

$$f(x_1) + f(x_2) = 6 \left[\frac{1}{21-5\sqrt{3}} + \frac{1}{21+5\sqrt{3}} \right] = 6 \cdot \frac{2 \cdot 21}{366} = \frac{252}{366} = \frac{42}{61}.$$

$$\text{Aproximación} = \frac{1}{2} \cdot \frac{42}{61} = \boxed{\frac{21}{61} \approx 0.34426}.$$

Comparación con el valor exacto:

$$I = \int_0^1 \frac{dx}{5x+1} = \frac{1}{5} \left[\ln(5x+1) \right]_0^1 = \frac{\ln 6}{5} \approx 0.35835.$$

El error absoluto es ≈ 0.0141 , razonable para una regla exacta solo hasta grado 3 aplicada a una función racional (no polinómica).

ENUNCIADO — PROBLEMA 5

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Problema 5 — BVP $-u'' + e^x u = \tan(x)$, $u(0) = u(1) = 0$

(a) **Diferencias finitas.** Malla uniforme $x_i = ih$, $h = 1/N$, $i = 0, \dots, N$, nodos interiores $i = 1, \dots, N-1$, $u_0 = 0$, $u_N = 0$ conocidos. Diferencia central:

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} + e^{x_i} u_i = \tan(x_i) \implies -u_{i-1} + (2 + h^2 e^{x_i}) u_i - u_{i+1} = h^2 \tan(x_i).$$

Matriz tridiagonal simétrica, con **diagonal variable** (a diferencia de un coeficiente constante):

$$A_h = \begin{pmatrix} 2 + h^2 e^{x_1} & -1 & & & \\ -1 & 2 + h^2 e^{x_2} & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 + h^2 e^{x_{N-1}} & \end{pmatrix}, \quad \mathbf{f} = \left(h^2 \tan(x_i) \right)_{i=1}^{N-1}.$$

Como $e^{x_i} > 0$, A_h sigue siendo estrictamente diagonalmente dominante (diagonal > 2), luego SPD e invertible.

Si $u(1) = \alpha \neq 0$: cambia solo la **última** ecuación ($i = N-1$), donde $u_N = \alpha$ se traslada al lado derecho:

$$-u_{N-2} + (2 + h^2 e^{x_{N-1}}) u_{N-1} = h^2 \tan(x_{N-1}) + \alpha.$$

La matriz A_h no cambia; solo la última componente de $\mathbf{f}$ se modifica sumándole α .

(b) **Elementos finitos.** Forma débil (multiplicar por $v \in H_0^1(0, 1)$ e integrar por partes, usando $v(0) = v(1) = 0$):

$$\int_0^1 u' v' dx + \int_0^1 e^x u v dx = \int_0^1 \tan(x) v dx, \quad \forall v \in H_0^1(0, 1).$$

Con la base de funciones sombrero $\{\varphi_i\}_{i=1}^{N-1}$ en la malla equidistante, $u_h = \sum_j c_j \varphi_j$, el sistema de Galerkin es

$$(K + M_{e^x}) \mathbf{c} = \mathbf{F}, \quad K_{ij} = \int_0^1 \varphi_i' \varphi_j' dx \text{ (rigidez estándar)}, \quad (M_{e^x})_{ij} = \int_0^1 e^x \varphi_i \varphi_j dx, \quad F_i = \int_0^1 \tan(x) \varphi_i dx.$$

K es la matriz de rigidez tridiagonal usual $\frac{1}{h}$ tridiag($-1, 2, -1$); M_{e^x} es una matriz de masa "ponderada" (tridiagonal, con entradas que en la práctica se aproximan por cuadratura sobre cada elemento, p.ej. regla del punto medio o trapecio, ya que e^x no es constante). El lado derecho $\mathbf{F}$ también se calcula por cuadratura numérica sobre cada elemento.

Tasa de convergencia: el problema es uniformemente elíptico (coeficiente $e^x \geq 1 > 0$ en $[0, 1]$) y la forma bilineal es coerciva y acotada en $H_0^1(0, 1)$; con $u \in H^2(0, 1)$ (dato $\tan(x)$ suave en $[0, 1]$, ya que $1 < \pi/2$), el lema de Céa y la estimación de interpolación lineal dan $\|u - u_h\|_{H^1} = O(h)$, y por dualidad de Aubin-Nitsche, $\|u - u_h\|_{L^2} = O(h^2)$.



ENUNCIADO — PROBLEMA 6

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Problema 6 — Iteración de potencias con eigenvalores complejos

$$M = \begin{pmatrix} \sqrt{3} & 0 & -\sqrt{3} \\ 0 & a & 0 \\ \sqrt{3} & 0 & \sqrt{3} \end{pmatrix}.$$

Espectro de M . El vector $e_2 = (0, 1, 0)^T$ es invariante: $Me_2 = ae_2$, con eigenvalor a . En el bloque 2×2 restante (coordenadas 1, 3),

$$B = \begin{pmatrix} \sqrt{3} & -\sqrt{3} \\ \sqrt{3} & \sqrt{3} \end{pmatrix}, \quad \text{tr } B = 2\sqrt{3}, \quad \det B = 3 - (-3) = 6.$$

Los eigenvalores de B resuelven $\lambda^2 - 2\sqrt{3}\lambda + 6 = 0$:

$$\lambda = \frac{2\sqrt{3} \pm \sqrt{12 - 24}}{2} = \sqrt{3} \pm i\sqrt{3} = \sqrt{3}(1 \pm i), \quad |\lambda| = \sqrt{3} \cdot \sqrt{2} = \sqrt{6}.$$

Así, el espectro completo de M es $\{a, \sqrt{3}(1+i), \sqrt{3}(1-i)\}$, con el par complejo conjugado de módulo común $\sqrt{6} \approx 2.449$. Geométricamente, B actúa en el plano (x, z) como una rotación de 45° combinada con una dilatación por $\sqrt{6}$ (ya que $1 \pm i = \sqrt{2} e^{\pm i\pi/4}$).

Caso $a = 1$: por qué falla la convergencia. Aquí $|a| = 1 < \sqrt{6}$, así que el eigenvalor a **no** es dominante — los eigenvalores de módulo máximo son el par complejo $\sqrt{3}(1 \pm i)$, ambos con el **mismo módulo** $\sqrt{6}$ (por ser conjugados). Al no existir un único eigenvalor real dominante, sino un par complejo con módulo empatado, la iteración de potencias normalizada, para cualquier w_0 con componente no nula en el subespacio invariante generado por el plano (x, z) (es decir, para casi todo $w_0 \in \mathbb{R}^3$, excluyendo solo el eje e_2 , de medida cero), produce iterados que **rotan indefinidamente** dentro de ese plano en incrementos angulares de 45° tras normalizar — la dirección de $w_k/\|w_k\|$ nunca se estabiliza, sino que cicla (con período 8 en el ángulo, ya que $8 \times 45^\circ = 360^\circ$). Por eso la iteración de potencias normalizada **no converge** para casi ningún vector inicial cuando $a = 1$.

Condición para convergencia y límite. Para que la iteración de potencias normalizada converja (para casi todo w_0) se necesita un eigenvalor **real y estrictamente dominante** en módulo sobre todos los demás. Como los otros dos eigenvalores tienen módulo fijo $\sqrt{6}$, esto requiere

$$|a| > \sqrt{6}.$$

En ese caso, escribiendo $w_0 = c_1 e_2 + (\text{componente en el plano } (x, z))$, la componente en el plano decae geométricamente (factor $(\sqrt{6}/|a|)^k \rightarrow 0$) frente a la componente en e_2 (que crece como a^k), y

$$\lim_{k \rightarrow \infty} \frac{w_k}{\|w_k\|} = \pm e_2 = \pm(0, 1, 0)^T$$

(con signo fijo si $a > 0$; alternando en cada paso si $a < 0$, exactamente como en el Problema 3 del examen de agosto de 2018). ■

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5. Consider the two-point boundary value problem:

$$-u'' + \exp(x)u = \tan(x) \quad \text{in } (0, 1), \quad \text{with } u(0) = u(1) = 0. \quad (1)$$

- (a) Write the finite difference approximation of (1) using *centered* differences on a uniform mesh with meshsize h . Write the matrix of the system and examine how the equations would change if $u(1) = \alpha \neq 0$.
- (b) Set up the finite element approximation for this problem, based on piecewise linear elements in equidistant points. State the convergence rate in an appropriate norm.
6. Consider the matrix

$$\begin{pmatrix} \sqrt{3} & 0 & -\sqrt{3} \\ 0 & a & 0 \\ \sqrt{3} & 0 & \sqrt{3} \end{pmatrix}$$

Explain why normalized power iteration fails to converge for almost every initial vector in $\mathbb{R}^3$ when $a = 1$. For what values of a will normalized power iteration converge for almost every vector in $\mathbb{R}^3$? What will it converge to?

7. Let h be the step size used for interpolating the function $f(x) = \sin(x)$ in $\left[0, \frac{\pi}{4}\right]$ and let $p(x)$ be the quadratic (three nodes) Lagrange interpolating polynomial using equally spaced nodes. Show that $|f(x) - p(x)| \leq \frac{h^3}{9\sqrt{3}}$.

Problema 7 — Cota de error de interpolación cuadrática para $\sin(x)$

Tres nodos equidistantes en $[0, \frac{\pi}{4}]$: $x_0 = 0$, $x_1 = h$, $x_2 = 2h$, con $2h = \pi/4$ (paso h). Sea $p(x)$ el interpolante de Lagrange de grado ≤ 2 . Como $f = \sin \in C^3$, la fórmula del error de interpolación polinómica da, para cada $x \in [x_0, x_2]$, la existencia de $\xi = \xi(x) \in (x_0, x_2)$ tal que

$$f(x) - p(x) = \frac{f'''(\xi)}{3!} (x - x_0)(x - x_1)(x - x_2).$$

Paso 1: acotar $|f'''|$. $f'''(x) = -\cos(x)$ para $f = \sin$. En $[0, \pi/4]$, $|\cos(x)|$ alcanza su máximo en $x = 0$: $|f'''(x)| \leq |\cos(0)| = 1 =: M_3$.

Paso 2: acotar el producto $(x - x_0)(x - x_1)(x - x_2)$. Con el cambio de variable $t = x - x_1 \in [-h, h]$ (centrado en el nodo medio), y usando $x - x_0 = t + h$, $x - x_2 = t - h$:

$$(x - x_0)(x - x_1)(x - x_2) = (t + h)t(t - h) = t(t^2 - h^2) = t^3 - h^2t.$$

Para $t \in [0, h]$ (por (im)paridad, basta considerar $t \geq 0$, ya que la función $g(t) = t(h^2 - t^2)$ es impar y su máximo en valor absoluto ocurre en $|t|$), definimos $g(t) = t(h^2 - t^2) = h^2t - t^3 \geq 0$.

Maximizamos:

$$g'(t) = h^2 - 3t^2 = 0 \implies t^* = \frac{h}{\sqrt{3}} \in (0, h).$$

$$g(t^*) = \frac{h}{\sqrt{3}} \left(h^2 - \frac{h^2}{3} \right) = \frac{h}{\sqrt{3}} \cdot \frac{2h^2}{3} = \frac{2h^3}{3\sqrt{3}}.$$

(Se verifica que es máximo: $g''(t) = -6t < 0$ en $t^* > 0$.) Así,

$$\max_{x \in [x_0, x_2]} |(x - x_0)(x - x_1)(x - x_2)| = \frac{2h^3}{3\sqrt{3}}.$$

Paso 3: combinar.

$$|f(x) - p(x)| \leq \frac{M_3}{3!} \cdot \frac{2h^3}{3\sqrt{3}} = \frac{1}{6} \cdot \frac{2h^3}{3\sqrt{3}} = \frac{h^3}{9\sqrt{3}}.$$

$$\boxed{|f(x) - p(x)| \leq \frac{h^3}{9\sqrt{3}} \quad \forall x \in \left[0, \frac{\pi}{4}\right].} \quad \blacksquare$$

Verificación numérica de orden de magnitud: $h = \pi/8 \approx 0.3927$, cota $\approx (0.3927)^3/(9\sqrt{3}) \approx 0.0606/15.588 \approx 0.00389$, un error diminuto consistente con la suavidad de $\sin(x)$ y el pequeño intervalo.